WORKED SOLUTION

STEP I 2017, Question 6

Pure • Integrals and Sign Changes

STEP I 2017, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q6

The key idea

A continuous function that never changes sign, and isn't identically zero, has a non-zero integral: it stays away from zero on some small interval, and that interval alone contributes something. Part (i) makes this precise, and parts (ii) and (iii) apply it to cleverly chosen functions. In (ii) the integrand $(x-\alpha)^2 g(x)$ has the same sign as g; in (iii) integration by parts turns the data about h into the conditions of (ii) for h'.

Solution

Part (i)

Suppose, for contradiction, that f does *not* take both positive and negative values on $[0,1]$. Then either $f(x) \geq 0$ for all x in $[0,1]$, or $f(x) \leq 0$ for all such x. Replacing f by $-f$ if necessary (this changes neither the hypothesis $\int_0^1 f(x)\,\mathrm{d}x = 0$ nor the conclusion), we may assume $f(x) \geq 0$ throughout.

We are told $f(c) \neq 0$ for some $c \in [0,1]$, so $f(c) > 0$. Since f is continuous at c, there is a $\delta > 0$ such that $f(x) > \frac{1}{2} f(c)$ for all x in $[0,1]$ with $|x-c| < \delta$. This set contains an interval $[p,q] \subseteq [0,1]$ of positive length $q-p$ (take $[c,\, c+\eta]$ if $c \leq \frac{1}{2}$ and $[c-\eta,\, c]$ if $c > \frac{1}{2}$, where $0 < \eta < \delta$ and $\eta \leq \frac{1}{2}$). Since $f \geq 0$ everywhere,

$$\int_0^1 f(x)\,\mathrm{d}x \;\geq\; \int_p^q f(x)\,\mathrm{d}x \;\geq\; \tfrac{1}{2} f(c)\,(q-p) \;>\; 0,$$

which contradicts $\int_0^1 f(x)\,\mathrm{d}x = 0$. Hence f takes both positive and negative values on $[0,1]$.

Ask yourself Why isn't "$f \geq 0$ and $f(c) > 0$, so the integral is positive" enough on its own?

A function can be positive at a single point without its integral noticing; a single point

has no width. It's continuity that spreads the positivity at c over an interval of positive length, and that is what forces the integral to be positive.

Part (ii)

Expanding the square and using the three conditions $(*)$,

$$\begin{aligned}\int_0^1 (x-\alpha)^2 g(x)\,\mathrm{d}x &= \int_0^1 x^2 g(x)\,\mathrm{d}x - 2\alpha\int_0^1 x g(x)\,\mathrm{d}x + \alpha^2\int_0^1 g(x)\,\mathrm{d}x \\ &= \alpha^2 - 2\alpha\cdot\alpha + \alpha^2\cdot 1 = 0.\end{aligned}$$

Now suppose, for contradiction, that $g(x)\neq 0$ for every $x\in[0,1]$. Then g cannot take both signs on $[0,1]$ (otherwise, by the assumption at the start of the question, it would vanish somewhere), so g is strictly positive throughout or strictly negative throughout.

Let $f(x)=(x-\alpha)^2 g(x)$, which is continuous. Since $(x-\alpha)^2\geq 0$, f has the same sign as g wherever it is non-zero, so f does not take both signs. But f is non-zero somewhere in $[0,1]$: $(x-\alpha)^2$ vanishes only at $x=\alpha$, so $f(x)\neq 0$ for every $x\in[0,1]$ other than (possibly) α. And $\int_0^1 f(x)\,\mathrm{d}x = 0$. By part (i), f must take both signs, a contradiction. Hence $g(x)=0$ for some x in $[0,1]$.

A linear example. With $g(x)=a+bx$ the conditions $(*)$ read

$$a+\tfrac{1}{2}b=1,\qquad \tfrac{1}{2}a+\tfrac{1}{3}b=\alpha,\qquad \tfrac{1}{3}a+\tfrac{1}{4}b=\alpha^2.$$

From the first, $a=1-\frac{1}{2}b$; substituting into the second gives $\frac{1}{2}+\frac{1}{12}b=\alpha$, so

$$b=12\alpha-6,\qquad a=4-6\alpha.$$

The third condition then becomes

$$\tfrac{1}{3}(4-6\alpha)+\tfrac{1}{4}(12\alpha-6)=\alpha-\tfrac{1}{6}=\alpha^2,$$

that is, $6\alpha^2-6\alpha+1=0$, so

$$\alpha=\frac{3\pm\sqrt{3}}{6}.$$

Notice that α isn't free: a linear g only works for these two values. Taking $\alpha=\frac{3+\sqrt{3}}{6}$ gives $b=2\sqrt{3}$ and $a=1-\sqrt{3}$, so

$$g(x)=1-\sqrt{3}+2\sqrt{3}\,x,$$

which is zero at

$$x=\frac{\sqrt{3}-1}{2\sqrt{3}}=\frac{3-\sqrt{3}}{6}\approx 0.211,$$

and this lies in $[0,1]$, as part (ii) says it must. (The other choice, $\alpha=\frac{3-\sqrt{3}}{6}$, gives $g(x)=1+\sqrt{3}-2\sqrt{3}\,x$, which is zero at $x=\frac{3+\sqrt{3}}{6}\approx 0.789$, also in $[0,1]$.)

Part (iii)

We show that $g = h'$ satisfies $(*)$ with $\alpha = 1 - \beta$. Here h' is continuous, as part (ii) requires. First,

$$\int_0^1 h'(x)\,\mathrm{d}x = h(1) - h(0) = 1.$$

Integrating by parts with $u = x$, $\mathrm{d}v = h'(x)\,\mathrm{d}x$,

$$\int_0^1 xh'(x)\,\mathrm{d}x = \Big[xh(x)\Big]_0^1 - \int_0^1 h(x)\,\mathrm{d}x = 1 - \beta.$$

Integrating by parts again, now with $u = x^2$,

$$\begin{aligned}\int_0^1 x^2h'(x)\,\mathrm{d}x &= \Big[x^2h(x)\Big]_0^1 - 2\int_0^1 xh(x)\,\mathrm{d}x \\ &= 1 - \beta(2-\beta) = 1 - 2\beta + \beta^2 = (1-\beta)^2.\end{aligned}$$

Ask yourself Where does the strange value $\frac{1}{2}\beta(2-\beta)$ come from?

It's chosen precisely so that the third integral comes out as the square of the second. After the two integrations by parts, the second integral is $1 - \beta$, so we need the third to be $(1-\beta)^2$, and that is exactly what $\frac{1}{2}\beta(2-\beta)$ delivers.

So h' is continuous and satisfies $(*)$ with $\alpha = 1 - \beta$. By part (ii), $h'(x) = 0$ for some x in $[0, 1]$.

Checks

- For $g(x) = 1 - \sqrt{3} + 2\sqrt{3}\,x$, direct integration gives $\int_0^1 g = 1$, $\int_0^1 xg = \frac{3+\sqrt{3}}{6} \approx 0.7887$ and $\int_0^1 x^2 g \approx 0.6220 = (0.7887)^2$.
- Part (iii) can be tested on $h(x) = (1 - \sqrt{3})x + \sqrt{3}\,x^2$, whose derivative is the g above. Then $h(0) = 0$, $h(1) = 1$, $\beta = \frac{3-\sqrt{3}}{6} \approx 0.2113$, and $\int_0^1 xh = \frac{1}{2}\beta(2-\beta) \approx 0.1890$. Its derivative vanishes at $x \approx 0.211$, so h has a turning point inside $[0, 1]$, as predicted.

Common slips

- In (i), arguing "$f \geq 0$, so $\int f > 0$" without using continuity. A non-negative function can have zero integral without being zero everywhere (if it is non-zero at only one point), so continuity is essential.
- In (ii), applying (i) to g itself. We know nothing about $\int g$ except that it is 1; part (i) has to be applied to $(x - \alpha)^2 g$, and we must check that this function is non-zero somewhere.
- In (iii), forgetting that h' must be continuous for (ii) to apply. That's why the question says h has a *continuous* derivative.

Beyond STEP

The conditions $(*)$ say that if g were a probability density, its mean would be α and its variance $\int (x-\alpha)^2 g = 0$. A genuine density with zero variance is impossible, and that's why g must dip below zero somewhere. Part (i) is a standard lemma of real analysis: a continuous, non-negative function with zero integral vanishes identically. It is used constantly, for example in the calculus of variations. Abbott's *Understanding Analysis* develops continuity and the Riemann integral carefully enough to prove it, and to prove the intermediate value theorem that the question lets you assume.

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